

# ASHVINY ARUMUGAM

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## PROFILE:

I am a professional Electronics Engineer with over 2 years of experience in Advantest Verigy 93K tester platform using Advantest Smartest 8 in semiconductor industry. I had the opportunity to drive several projects which helped me to enhance both technical and managerial skills. I am looking forward to continuing my career in an organization that utilizes my knowledge and experience.

## WORK EXPERIENCE:

### Validation Engineer I(Silicon), Tessolve Ltd (Qualcomm consultant)

Aug 2021 – May 2023

- Assisted in the development of test program using 93K tester with Smartest 8 software.
- Worked in improving the yield and reducing test time for improvement in quality of testing.
- Contributed for testing the vectors at Wafer level and Final Package level.
- Tracking the updates and changes in test program using SVN and GIT.
- Involved in using statistical data analysis tools to analyse data for volume parts.
- Supported in debugging activities across parameters like process, voltage, and temperature in production.
- Performed development of test program and debugging of structural blocks (ATPG, TDF and MBIST)
- Communicated and collaborated with cross functional teams to deliver accurate validation results.

### Learnings through Advantest 93K

- Key concepts and components of V93K system.
- Understanding of SmarTest8 SW concepts and how to use them.
- Setting up of test program using SmarTest8 features.
- Pin configuration setup of levels, timings, and vectors
- Usage of shmoo tools and test tables

## Technical Skills

| Category                        | Tools and Technologies                                     |
|---------------------------------|------------------------------------------------------------|
| Validation tools                | Advantest 93K (Automated Test Equipment), Exensio, O Plus  |
| Database Management tools       | MySQL (Query writing, Data modelling, Database Management) |
| Data Visualization              | Power BI (Dashboards, Reports and Data insights), MS Excel |
| Programming/Scripting Languages | Python, C                                                  |
| Operating systems               | Microsoft Windows, Linux (Red Hat CentOS)                  |

## EDUCATION

- **Nanyang Technological University**, Singapore

Aug 2020 – Aug 2021

*Master of Science, Electronics Engineering*

- **Anna University - MIT**, Chennai, India

July 2015 – May 2019

*Bachelor of Engineering, Electronics and Instrumentation*

## **PROJECTS:**

### **Design & implementation of JFET based colour sensors**

**Nanyang Technological University, Singapore**

**Oct 2020 – Jun 2021**

- Examination and realization of optical and colour sensing in silicon.
- Analysing the behavioural changes in JFET when used as photosensor.
- Calculation of certain optical characteristics namely optical conductivity, absorption coefficient, full width half maximum, attenuation loss and resistivity in order to analyse its sensing performance.
- In addition, theoretical calculations for the fabrication of proposed structure were made.

### **Three axis control of a pen using PLC.**

**Anna University-MIT, India**

**Aug 2018 – April 2019**

- Design a three-axis control system to control the robotic hand containing a pen which could draw on a sheet of paper.
- The entire system is controlled by MITSUBISHI-PLC and the software used was GX Developer.

### **Detector Wireless controlled bot using Arduino**

**Anna University-MIT, India**

**Aug 2017 – Feb 2018**

- Worked in building DTMF (Dual tone multi frequency) controlled bot to capture live footage of the environment using a mobile phone or laptops which would travel in places where human intervention is risky.
- DTMF controlled module: MT8870DE
- Microcontroller platform: Arduino.

## **WORKSHOPS AND TRAININGS**

- Control system training in Salem Steel Plant, India, 2016.
- Security system training in Chennai International airport, India, 2017.
- PIC-Embedded Workshop on MSP430, Anna University, India. 2018

## **INTERN EXPERIENCE:**

### **Smart Farm System Laboratory**

**Gyeongsang National University, Jinju, South Korea**

**Aug 2019 – Oct 2019**

- Acquisition and Integration of data
- Responsible to acquire and monitor data from various sensors using micro controller and analysis of acquired data at regular intervals. Collection of data (like soil temperature, minimum and maximum leaf temperature, leaf breadth and plant height) from strawberry plant in three different farms.
- PLC used – MOACON PLC.
- Software used – MOACON STUDIO.